King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 243

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Using Newton's Method to approximate the zero(s) of the function $f(x) = x - 3 + \ln x$, we get the iterative formula

(a) $x_{n+1} = \dfrac{x_n^2 - x_n + x_n \ln x_n}{x_n + 1}$

(b) $x_{n+1} = \dfrac{-2x_n - x_n \ln x_n}{x_n + 1}$

(c) $x_{n+1} = \dfrac{4x_n - x_n \ln x_n}{x_n + 1}$

(d) $x_{n+1} = \dfrac{4x_n + 2x_n \ln x_n}{x_n + 1}$

(e) $x_{n+1} = \dfrac{2x_n^2 - 2x_n + x_n \ln x_n}{x_n + 1}$

2. Let f be a differentiable function and have an inverse on $(-\infty, \infty)$. Assume that $f(2) = 6$, $f'(2) = 4$, $f(6) = 1$, and $f'(6) = 2$. Then $(f^{-1})'(6) =$

(a) $\dfrac{2}{3}$

(b) $\dfrac{1}{4}$

(c) $\dfrac{1}{2}$

(d) 1

(e) $\dfrac{1}{6}$

3. If $y = \dfrac{x^3+3}{x^2+2}$, then $\dfrac{dy}{dx} =$

(a) $\dfrac{3x^2-2x}{(x^2+2)^2}$

(b) $\dfrac{-x^3+x^2-1}{(x^2+2)^2}$

(c) $\dfrac{x^4+3x^2-2x}{(x^2+2)^2}$

(d) $\dfrac{x^3-x^2+1}{x^2+2}$

(e) $\dfrac{x^4+6x^2-6x}{(x^2+2)^2}$

4. All edges of a cube are expanding at a rate of 3 cm/s. How fast is the volume changing when each edge is 2 cm?

(a) 12 cm^3/s

(b) 27 cm^3/s

(c) 36 cm^3/s

(d) 9 cm^3/s

(e) 18 cm^3/s

5. If (a, b), where $a > 1$, is the point on the graph of $f(x) = \dfrac{x+2}{x-2}$ at which the tangent line is parallel to the line $4x + y = 1$, then $4a + b =$

(a) -2

(b) -8

(c) 1

(d) 17

(e) 7

6. If $f(x) = \dfrac{6}{4x+2}$, then $f^{(100)}(1) =$

(a) $-\left(\dfrac{2}{3}\right)^{100} \cdot 100!$

(b) $\dfrac{2^{100}}{3^{101}} \cdot 100!$

(c) $-\dfrac{2^{100}}{3^{99}} \cdot 99!$

(d) $\left(\dfrac{2}{3}\right)^{100} \cdot 100!$

(e) $6 \cdot \left(\dfrac{2}{3}\right)^{100} \cdot 100!$

7. If $y = Ax + B$ is an equation of the tangent line to the graph of $f(x) = 2 + e^x \tan x$ at the point $(0, 2)$, then $A^2 + B^2 =$

(a) 5

(b) 2

(c) 4

(d) 8

(e) 1

8. If $y = \dfrac{(2+\sqrt{x})(3+\sqrt[3]{x})}{(4+\sqrt[4]{x})(5+\sqrt[5]{x})}$, then the value of $\dfrac{dy}{dx}$ at $x = 1$ is equal to

(a) $\dfrac{1}{15}$

(b) $\dfrac{5}{3}$

(c) $\dfrac{2}{5}$

(d) $\dfrac{7}{30}$

(e) $\dfrac{2}{15}$

9. If $f(x) = x\tan^{-1}\left(\frac{x}{2}\right) - \ln(4+x^2)$, then $f'(x) =$

(a) $x\tan^{-1}\left(\frac{x}{2}\right) + \frac{1}{4+x^2}$

(b) $\tan^{-1}\left(\frac{x}{2}\right) - \frac{x}{4+x^2}$

(c) $\tan^{-1}\left(\frac{x}{2}\right)$

(d) $\tan^{-1}\left(\frac{x}{2}\right) + \frac{4x}{4+x^2}$

(e) $x\tan^{-1}\left(\frac{x}{2}\right) + \frac{4x}{4+x^2}$

10. Let $f(1) = -3$, $f'(1) = 6$, $g(2) = 2$, and $g'(2) = 4$. If $h(x) = f\left(\frac{x}{g(x)}\right)$, then $h'(2) =$

(a) -9

(b) -3

(c) $-\frac{3}{2}$

(d) 27

(e) $\frac{9}{2}$

11. The slope of the tangent line to the curve

$$2 + \tan(xy) = x^2 + y^2 + \tan(1)$$

at the point $(1, 1)$ is equal to

(a) $\frac{\sec^2 1}{2 - \sec^2 1}$

(b) $\frac{2 - \sec^2 1}{1 - \sec^2 1}$

(c) 2

(d) -1

(e) 0

12. The value(s) of c satisfying the conclusion of the Mean Value Theorem when applied to $f(x) = \sqrt{2-x}$ on the interval $[-7, 2]$ is (are)

(a) $-\frac{1}{2}$

(b) -2 and -4

(c) $-\frac{1}{4}$ and $\frac{1}{4}$

(d) $-\frac{2}{3}$

(e) $-\frac{1}{4}$

13. The maximum value of $f(x) = \dfrac{x}{(x^2+1)^2}$ on $[-1, 1]$ is equal to

(a) $\frac{1}{4}$

(b) $\frac{3\sqrt{3}}{8}$

(c) 0

(d) $\frac{\sqrt{3}}{4}$

(e) $\frac{3\sqrt{3}}{16}$

14. If $f(x) = \left(1 + \dfrac{1}{x}\right)^x$, then $f'(1) =$

(a) $-\frac{1}{4} + \frac{1}{2}\ln 2$

(b) $2 + 2\ln 2$

(c) $1 + 2\ln 2$

(d) $-1 + 2\ln 2$

(e) $2 - \ln 2$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$f' = \frac{x+1}{x}$, so $x_{n+1} = x_n - \frac{x_n(x_n-3+\ln x_n)}{x_n+1} = \frac{4x_n - x_n \ln x_n}{x_n+1}$.
2	**(b)**	$f^{-1}(6) = 2$, so $(f^{-1})'(6) = 1/f'(2) = 1/4$.
3	**(e)**	Quotient rule: numerator $3x^2(x^2+2) - 2x(x^3+3) = x^4 + 6x^2 - 6x$.
4	**(c)**	$V = s^3$, $V' = 3s^2 s' = 3(4)(3) = 36$ cm^3/s.
5	**(d)**	Slope -4: $(x-2)^2 = 1$, $a = 3$, $b = 5$, so $4a + b = 17$.
6	**(d)**	$f^{(n)} = 6(-1)^n n!\, 4^n (4x+2)^{-(n+1)}$; at $x = 1$, $n = 100$: $(2/3)^{100} \cdot 100!$.
7	**(a)**	$f'(0) = 1$, line $y = x + 2$, so $A^2 + B^2 = 5$.
8	**(a)**	Log differentiation: $y(1) = \frac{2}{5}$, $(\ln y)'(1) = \frac{1}{6}$, so $y'(1) = \frac{1}{15}$.
9	**(c)**	The $\frac{2x}{4+x^2}$ terms cancel, leaving $f'(x) = \tan^{-1}(x/2)$.
10	**(a)**	$h'(2) = f'(1) \cdot \frac{g(2) - 2g'(2)}{g(2)^2} = 6 \cdot (-\frac{3}{2}) = -9$.
11	**(d)**	$\sec^2 1(1 + y') = 2 + 2y'$ gives $y'(\sec^2 1 - 2) = -(\sec^2 1 - 2)$, so $y' = -1$.
12	**(e)**	Average slope $-\frac{1}{3}$; $\frac{-1}{2\sqrt{2-c}} = -\frac{1}{3}$ gives $c = -\frac{1}{4}$.
13	**(e)**	Critical points $x = \pm\frac{1}{\sqrt{3}}$; $f(\frac{1}{\sqrt{3}}) = \frac{3\sqrt{3}}{16} > f(1) = \frac{1}{4}$.
14	**(d)**	$(\ln f)' = \ln(1 + \frac{1}{x}) - \frac{1}{x+1}$; at 1: $\ln 2 - \frac{1}{2}$, times $f(1) = 2$.